

COVER LETTER

Date: August 11, 2026

Dear Hiring Manager,

I am writing to express my enthusiastic interest in AI/ML Engineer role at your organization. As a 2026 B.Tech undergraduate at IIT Jodhpur with a Minor in Artificial Intelligence & Data Engineering, I bring hands-on experience across the entire lifecycle of AI systems including dataset curation, model training, fine-tuning, backend development, high-throughput deployment, and real-time monitoring.

Across my industry internships and research environments, I have consistently turned core ML concepts into scalable, production-grade applications. At Thuriyam AI, I engineered automated transcription and NLP analysis pipelines using FastAPI and Gemini models processing 10,000+ daily call recordings. To optimize operational overhead, I developed a deterministic subword pre-filtering engine that reduced LLM API costs by 97% while maintaining >99.98% system reliability across 300,000+ monthly requests. At an AI Stealth Startup, I curated a 150K-sample multi-label dataset to train a BERT-based moderation model that achieved 95.8% accuracy with sub-50 ms latency, over 95% faster than heavy LLM guardrails. Additionally, I architected end-to-end speech-to-speech conversational agents incorporating streaming and caching to achieve sub-1-second total response latency.

Complementing my engineering background, I have a strong foundation in NLP research and data science. During my research at the IISc Bangalore NLP Lab, I built cross-lingual embedding datasets using NV-Retriever hard negative mining, achieving a 5% benchmark improvement on Semantic Textual Similarity (STS) via soft-prompt tuning on e5-Mistral-7B with only 0.001% trainable parameters. Furthermore, my predictive modeling work with PySpark, Pandas, and tree-based ensembles (XGBoost, LightGBM, SHAP) secured 6th place out of 23 IITs at the Inter-IIT Tech Meet 13.0.

Whether the role demands fine-tuning open-source LLMs, building agentic workflows, curating specialized datasets, or deploying low-latency backend microservices, I thrive in fast-paced teams driving cutting-edge AI. Thank you for your time and consideration, and I look forward to discussing how my skills can contribute to your team.

Yours Faithfully,

Yogesh Sharma

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